Understanding Influence Functions

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This note aims at providing insights and information on influence functions which are now at the core of semi-parametric statistics used in causal inference. The formalism here follows the line of Kennedy [6], which work presents a more in-depth introduction on such statistical tools.

As presented in Section 1, we consider a target parameter ψ, which is influence function is defined in Section 2. Based on such a function, statistical estimators are derived in Section 3 and further analyzed in Section 4. Eventually, we discuss in Section 5 the statistical efficiency of influence function based estimators.

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1 Problem Setting

1.1 Notation

We consider the standard setup for functional estimation problems, where we observe a sample of independent observations $(Z_1, \dots, Z_n) \in \mathcal{Z}$, each identically distributed according to some unknown probability distribution $\mathbb{P}$. This distribution is assumed to lie within a model, denoted as $\mathcal{P}$, representing a set of distributions.

Let $\mathcal{Z}$ be a compact metric space equipped with a measure, such as the Lebesgue measure, and let $\mathbb{P}$ be a measure over $\mathcal{Z}$ that is absolutely continuous with respect to the measure of $\mathcal{Z}$. We define $\mathcal{M}$ to be the set of all such measures whose Radon/Nikodym derivatives belong to $L_2(\mathcal{Z})$, denoted by $p \in L_2(\mathcal{Z})$. We write counterfactual outcomes as $Y^{(t)}$, i.e., the value of Y that would have been observed had we set $T = t$.

At times we subscript expectations and other quantities with the distribution under which they are taken, i.e., $\mathbb{E}_{\mathbb{P}}(Y \mid X = x)$ for an expectation under distribution $\mathbb{P}$. When the distribution is clear from context, we sometimes omit subscripts. We use $\mathbb{P}_n$ to denote the empirical measure so that sample averages are written as $\mathbb{P}_n(f) = \mathbb{P}_n\{f(Z)\} = \frac{1}{n}\sum_i f(Z_i)$. For a possibly random function f, we similarly write $\mathbb{P}(f) = \mathbb{P}\{f(Z)\} = \int f(z)d\mathbb{P}(z)$, and we let $\|f\|^2 = \int f(z)^2 d\mathbb{P}(z)$ denote the squared $L_2(\mathbb{P})$ norm.

1.2 Target Parameters/Functional

The goal in this setup is to estimate a structured combination of components known as a target parameter or functional,

$$\begin{aligned} \psi : \mathcal{M} &\longrightarrow \mathcal{D} \\ P &\longmapsto \psi(P) \end{aligned}$$

where $\mathcal{D}$ can be any function space. A functional can be viewed as a map from the model to some function space. Here, we will mostly focus on $\mathcal{D} = \mathbb{R}$. Here we give a list of examples of functionals, some of which arise from causal inference problems, and some of which do not.

Exemple 1. *(Regression function) Suppose $Z = (X, Y)$. The regression of Y on X is given by*

$$\psi = \psi(x) = \mathbb{E}(Y \mid X = x).$$

Note that in this case $\mathcal{D}$ is the set of function from $\mathbb{R}^d$ to $\mathbb{R}$.

Exemple 2. *(Average treatment effect) Suppose $Z = (X, T, Y)$ for confounders X, treatment T, and outcome Y. Then under causal positivity, consistency, and no unmeasured confounding assumptions, the mean outcome if all in the population were treated at level $T = t$ is identified as the expected regression function*

$$\psi = \mathbb{E}\left\{Y^{(t)}\right\} = \mathbb{E}\left\{\mu_t(X)\right\},$$

where we use the convention $\mathbb{E}(Y \mid X, T = t) = \mu_t(X)$ for $\mu_t(x) = \mathbb{E}(Y \mid X = x, T = t)$, so the outer expectation in the above is over the marginal distribution of X.

2 Influence functions

2.1 Von-Mises expansion

We suppose the functional $\psi : \mathcal{M} \mapsto \mathbb{R}$ is smooth which allows us to define its Gâteaux derivative:

Definition 2.1. *(Gateaux derivative) The map $\psi'(P, \cdot) : \mathcal{M} \to \mathbb{R}$ where $\psi'(P, H) = \frac{\partial \psi(P+tH)}{\partial t}\Big|_{t=0}$ is called the Gâteaux derivative at P if the derivative exists and is linear and continuous in H. We say ψ is Gâteaux differentiable at P if $\psi'(P, \cdot)$ exists at P.*

As a consequence, the functional can be written in a kind of *distributional Taylor expansion*

$$\psi(Q) = \psi(P) + \psi'(P, Q - P) + R_2(P, Q) \tag{1}$$

for distributions Q and P, often called a von Mises expansion [7], where $R_2(P, Q)$ is a second-order remainder term [6] (which means it only depends on products or squares of differences between Q and P). Intuitively, the von Mises expansion above is just an infinite-dimensional or distributional analog of a Taylor expansion, it describes how the functional ψ changes locally when the distribution changes from P to Q. This expansion allows to define the influence function of our target parameter.

Definition 2.2. *(Influence function) Let ψ be Gâteaux differentiable at P. A function $\varphi(\cdot; P) : \mathcal{X} \to \mathbb{R}$ is the influence function of ψ w.r.t the distribution P if it satisfies for all $Q \in \mathcal{M}$,*

$$\psi'(P, Q - P) = \int \varphi(z; P) dQ(z) \tag{2}$$

where $\varphi(z; P)$ is a mean-zero, finite-variance function satisfying $\int \varphi(z; P) dP(z) = 0$ and $\int \varphi(z; P)^2 dP(z) < \infty$.

The function φ has been referred to as an influence function [1, 4, 6], pathwise derivative gradient [8, 5], and Neyman orthogonal score [10, 3]. Note also that the existence and uniqueness of the influence function is guaranteed by the Riesz representation theorem, since the domain of ψ is a bijection of $L_2(\mathcal{Z})$ and consequently a Hilbert space.

Remark 3. *However it can be important to distinguish between the influence function for a parameter, as in* (2)*, and the influence function for an estimator. To distinguish between the two, we typically refer to the influence function for a parameter as in* (2) *as an influence curve.*

2.2 Pathwise differentiability

To derive influence functions, we introduce pathwise differentiability which account for fluctuations around the true measure more complicated nonparametric models is through a technical device called the parametric submodel [9].

Definition 2.3. *Parametric submodel A parametric submodel is a smooth parametric model $\mathcal{P}_\epsilon = \{P_\epsilon : \epsilon \in \mathbb{R}\}$ that satisfies (i) $\mathcal{P}_\epsilon \subseteq \mathcal{P}$, and (ii) $P_{\epsilon=0} = \mathbb{P}$.*

Thus, in words, a parametric submodel is a parametric model that (i) is contained in the larger model $\mathcal{P}$ of interest, and (ii) equals the true distribution at $\epsilon = 0$, i.e., contains the truth $\mathbb{P}$.

A common choice [4, 5, 6] of submodel for nonparametric $\mathcal{P}$ is, for some mean-zero function $h : \mathcal{Z} \to \mathbb{R}$,

$$p_\epsilon(z) = d\mathbb{P}(z)\{1 + \epsilon h(z)\} \tag{3}$$

where $\|h\|_\infty \leq M < \infty$ and $\epsilon < 1/M$ so that $p_\epsilon(z) \geq 0$. Note for this submodel the score function is $s_e(z) = \frac{\partial}{\partial \epsilon} \log p_\epsilon(z)\big|_{\epsilon=0} = \frac{\partial}{\partial \epsilon} \log\{1 + \epsilon h(z)\}\big|_{\epsilon=0} = h(z)$.

Using the Von-Mises expansion (1) taking $(P, Q) = (P_\epsilon, P)$, we have

$$\begin{aligned}
\psi(P_\epsilon) &= \psi(P) + \psi'(P_\epsilon - P; P) + R_2(P, P_\epsilon) \\
&= \psi(P) + \int \varphi(z; P) dP_\epsilon(z) + R_2(P, P_\epsilon) \\
&= \psi(P) + \epsilon \int \varphi(z; P) h(z) dP(z) + R_2(P, P_\epsilon).
\end{aligned}$$

Because $R_2(P, P_\epsilon)$ is second order (only depends on products or squares of differences between P_ϵ and P),

$$\frac{\partial}{\partial \epsilon} R_2(P, P_\epsilon)\bigg|_{\epsilon=0} = 0 \tag{4}$$

This condition above is essentially equivalent to what [3] refer to as Neyman orthogonality. Next, we get for every smooth submodel P_ϵ

$$\frac{\partial}{\partial \epsilon} \psi(P_\epsilon)\bigg|_{\epsilon=0} = \int \varphi(z; \mathbb{P}) h(z) dP(z)$$

This notion of smoothness is pathwise differentiability, i.e., that

$$\frac{\partial}{\partial \epsilon}\psi\left(P_\epsilon\right)\Big|_{\epsilon=0} = \int \varphi(z;\mathbb{P}) s_\epsilon(z) d\mathbb{P}(z) \tag{5}$$

Pathwise differentiability (5), and the von Mises expansion (2) more generally, play key roles in constructing estimators that attain the objectives on debiasing, multiply robustness and attain lower bounds.

2.3 Deriving the influence function

We now present two main ways of deriving the influence function.

2.3.1 Pathwise derivative

Using the pathwise derivative formula, we can compute the influence function of our functional.

Exemple 4. *(Regression) We consider the regression function* $\mathbb{E}(Y \mid X = x)$*. Here we will show that*

$$\varphi(z;P) = \frac{1(X=x)}{P(X=x)}\left[Y - \mathbb{E}_P(Y \mid X = x)\right]$$

is the efficient influence function when X *is discrete, by showing that* (2) *holds for this choice of* φ*. Let* $s_\epsilon(z) = \frac{\partial}{\partial \epsilon}\log dP_\epsilon(z)\big|_{\epsilon=0}$ *denote the submodel score, and note*

$$\begin{aligned}
\mathbb{E}\{s(Z) \mid X = x\} &= \int \frac{\partial}{\partial \epsilon}\log dP_\epsilon(z)\Big|_{\epsilon=0} d\mathbb{P}(y \mid x) \\
&= \int \frac{\partial}{\partial \epsilon}\log\left(P_\epsilon(X=x) dP_\epsilon(y \mid x)\right)\Big|_{\epsilon=0} d\mathbb{P}(y \mid x) \\
&= \int \left[\frac{\partial}{\partial \epsilon}\log P_\epsilon(X=x)\Big|_{\epsilon=0} + \frac{\partial}{\partial \epsilon}\log dP_\epsilon(y \mid x)\Big|_{\epsilon=0}\right] d\mathbb{P}(y \mid x) \\
&= \frac{\partial}{\partial \epsilon}\log P_\epsilon(X=x)\Big|_{\epsilon=0}
\end{aligned}$$

where the last equality uses the facts that $\int d\mathbb{P}(y \mid x) = 1$ *and that scores have mean zero, i.e.,*

$$\int \frac{\partial}{\partial \epsilon}\log dP_\epsilon(y \mid x)\Big|_{\epsilon=0} d\mathbb{P}(y \mid x) = 0$$

Thus in this case the pathwise derivative on the left-hand side of (2) *is*

$$\begin{aligned}
\frac{\partial}{\partial \epsilon}\int y dP_\epsilon(y \mid x)\Big|_{\epsilon=0} &= \int y\left(\frac{\partial}{\partial \epsilon}\log dP_\epsilon(y \mid x)\Big|_{\epsilon=0}\right) d\mathbb{P}(y \mid x) \\
&= \int y\left[\frac{\partial}{\partial \epsilon}\log \frac{dP_\epsilon(z)}{P_\epsilon(X=x)}\Big|_{\epsilon=0}\right] d\mathbb{P}(y \mid x) \\
&= \int y\left[\frac{\partial}{\partial \epsilon}\log dP_\epsilon(z) - \frac{\partial}{\partial \epsilon}\log P_\epsilon(X=x)\Big|_{\epsilon=0}\right] dP_\epsilon(y \mid x) \\
&= \mathbb{E}\left\{Y s_\epsilon(Z) \mid X = x\right\} - \mathbb{E}\left\{s_\epsilon(Z) \mid X = x\right\}\mathbb{E}(Y \mid X = x).
\end{aligned}$$

where the first equality holds as long as we can exchange integrals and derivatives and since $P_{\epsilon=0} = P$*. Now for the right-hand side of we have*

$$\begin{aligned}
\int \varphi(z;\mathbb{P}) s_\epsilon(z) d\mathbb{P}(z) &= \mathbb{E}\left[\frac{1(X=x)}{P(X=x)}[Y - \mathbb{E}(Y \mid X = x)] s_\epsilon(Z)\right] \\
&= \mathbb{E}\left\{Y s_\epsilon(Z) \mid X = x\right\} - \mathbb{E}\left\{s_\epsilon(Z) \mid X = x\right\}\mathbb{E}(Y \mid X = x)
\end{aligned}$$

by iterated expectation. This yields the result.

2.3.2 Practical tricks

We can use tricks to compute the influence function. To help make ideas concrete, we introduce an operator $\mathbb{IF} : \Psi \to L_2(P)$ that maps functionals $\psi : \mathcal{P} \to \mathbb{R}$ to their influence functions $\varphi(z) \in L_2(P)$ in a nonparametric model. Then the tricks that can be used sequentially are:

1. Pretend the data are discrete.
2. Treat influence functions as derivatives, allowing use of differentiation rules.
3. Use influence function building blocks, e.g.,

$$\mathbb{IF}(\mathbb{E}(Y \mid X = x)) = \frac{\mathbf{1}(X = x)}{P(X = x)}\{Y - \mathbb{E}(Y \mid X = x) \quad \mathbb{IF}(\mathbb{P}(X = x)) = \mathbf{1}(X = x) - \mathbb{P}(X = x)\}$$

a) (product rule)

$$\mathbb{IF}(\psi_1\psi_2) = \mathbb{IF}(\psi_1)\psi_2 + \psi_1\mathbb{IF}(\psi_2)$$

b) (chain rule)

$$\mathbb{IF}(f(\psi)) = f'(\psi)\mathbb{IF}(\psi)$$

Exemple 5. *(Average treatment effect) We have $\psi_t = \mathbb{E}\{\mu_t(X)\}$, let $\mu_t(x) = \mathbb{E}(Y \mid X = x, T = t)$, $\pi_t(x) = \mathbb{P}(T = t \mid X = x)$, and $p(x) = \mathbb{P}(X = x)$, then the influence function is given by*

$$\mathbb{I}(\psi_t) = \mathbb{IF}\left(\sum_x \mu_t(x)p(x)\right) = \sum_x [\mathbb{IF}\{\mu_t(x)\}p(x) + \mu(x)\mathbb{IF}\{p(x)\}]$$

$$= \sum_x \frac{\mathbf{1}(X = x, T = t)}{p(T = t, X = x)}(Y - \mu_t(x))p(x) + \mu_t(x)(\mathbf{1}(x = X) - p(x))$$

$$= \frac{\mathbf{1}(T = t)}{\pi_t(X)}(Y - \mu_t(X)) + \mu_t(X) - \psi_1$$

3 Building estimators

Our objective is to estimate $\psi(\mathbb{P})$. However, since we only have access to the empirical distribution $\mathbb{P}_n$, we can instead build an estimated distribution $\hat{\mathbb{P}}$ and estimate $\psi(\mathbb{P})$ through $\psi(\widehat{\mathbb{P}})$. The generic plug-in estimator is defined as:

$$\hat{\psi}_{pi} = \psi(\widehat{\mathbb{P}}) \tag{6}$$

The von Mises expansion allows us to express $\psi(\mathbb{P}) - \psi(\widehat{\mathbb{P}})$ as:

$$\psi(\widehat{\mathbb{P}}) - \psi(\mathbb{P}) = -\int \varphi(z; \widehat{\mathbb{P}})d\mathbb{P}(z) + R_2(\widehat{\mathbb{P}}, \mathbb{P})$$

This expansion suggests that the generic plug-in estimator $\psi(\widehat{\mathbb{P}})$ has a first order bias, being $\int \varphi(z; \widehat{\mathbb{P}})d\mathbb{P}(z)$.

Exemple 6. *(Average treatment effect) A plug-in estimator for the average treatment effect functional $\psi = \mathbb{E}\{\mathbb{E}(Y \mid X, T = t)\}$ is given by $\widehat{\psi}_{pi} = \mathbb{P}_n\{\widehat{\mu}_t(X)\}$, for $\widehat{\mu}_t(x)$ an estimator of $\mu_t(x) = \mathbb{E}(Y \mid X = x, T = t)$. Suppose for simplicity that $\widehat{\mu}_t$ is estimated on a separate sample independent of the sample on which $\mathbb{P}_n$ operates. Then the bias of this plug-in estimator is given by*

$$\mathbb{E}\left(\psi - \widehat{\psi}_{pi}\right) = \int \mathbb{E}\{\mu_t(x) - \widehat{\mu}_t(x)\}d\mathbb{P}(x)$$

which is just the integrated bias of the regression estimator $\widehat{\mu}_t$ itself.

The expansion (1) also suggests how to correct or de-bias generic plug-in estimators by working on the bias term $-\int \varphi(z,\widehat{\mathbb{P}})d\mathbb{P}(z)$.

Here we present the main strategies to do so:

- One step Correction: Estimate $\int \varphi(z,\widehat{\mathbb{P}})d\mathbb{P}(z)$ directly.
- Estimating Equation: Build an estimator ψ that eliminates the term $\int \varphi(z,\widehat{\mathbb{P}})d\mathbb{P}(z)$ to simplify and debias the estimation.
- Targeted learning: Use the plug-in estimator on nuisance parameters that eliminate the term $\int \varphi(z,\widehat{\mathbb{P}})d\mathbb{P}(z)$ to debias the estimation.

3.1 One-step estimator

The first estimator $\hat{\psi}^{(os)}$ is the one-step estimator or bias correction estimator [6], which consists in estimating the bias term $\int \varphi(z,\widehat{\mathbb{P}})d\mathbb{P}(z)$ and subtracting it off. Since this expression is just a mean, a natural estimator is given by the corresponding sample average $-\mathbb{P}_n\{\varphi(Z,\widehat{\mathbb{P}})\}$, leading to the one-step estimator

$$\hat{\psi}^{(os)} = \psi(\widehat{\mathbb{P}}) + \mathbb{P}_n\left\{\varphi\left(Z,\widehat{\mathbb{P}}\right)\right\}$$

Exemple 7. *(Average treatment effect) Noting $\psi = \mathbb{E}\{\mu_t(X)\}$ and we have from the calculations above:*

$$\varphi(Z;\mathbb{P}) = \frac{\mathbf{1}(T=t)}{\mathbb{P}(T=t \mid X)}(Y-\mu_t(X)) + \mu_t(X) - \psi$$

Therefore,

$$\begin{aligned}\hat{\psi}^{(os)} &= \hat{\psi}_{pi}(\widehat{\mathbb{P}}) + \mathbb{P}_n\left\{\varphi\left(Z,\widehat{\mathbb{P}}\right)\right\} \\ &= \sum_{i=1}^{n}\left(\hat{\mu}_t(X_i) + \mathbf{1}(T_i=t)\frac{Y_i-\hat{\mu}_t(X_i)}{\hat{\pi}_t(X_i)}\right)\end{aligned}$$

3.2 "Estimating Equation" technique

Another approach is commonly referred to as the estimating equation estimator [6]. This method works by identifying a value for the target parameter along with a set of nuisance parameters such that the estimated influence function directly satisfies the condition $\mathbb{P}_n\varphi\left(Z;\widehat{\mathbb{P}}\right) = 0$.

To apply the estimating equation approach, we assume that the influence function at any given distribution depends only on the target parameter ψ and a (potentially infinite) set of nuisance parameters, denoted η. Therefore, instead of expressing the influence function as $\varphi(Z;\mathbb{P})$ for the distribution $\mathbb{P}$, we rewrite it as $\varphi(Z;\psi,\eta)$. If the influence function can be structured in this way, the procedure involves first estimating $\hat{\eta}$ using a flexible and fast-converging method. After obtaining $\hat{\eta}$, we then determine a value for $\hat{\psi}^{(ee)}$ such that when it is substituted into the estimating equation, we have $\mathbb{P}_n\varphi\left(Z;\widehat{\mathbb{P}}\right) = \mathbb{P}_n\varphi\left(Z;\hat{\psi}^{(ee)},\hat{\eta}\right) = 0$.

Exemple 8. *(Average treatment effect) We have $\psi = \mathbb{E}\{\mu_t(X)\}$ and we have from the calculations above:*

$$\varphi(Z;\mathbb{P}) = \varphi(Z;\psi,\mu_t,\pi) = \mathbf{1}(T=t)\frac{(Y-\mu_t(X))}{\pi_t(X)} + \mu_t(X) - \psi$$

Therefore, setting $\mathbb{P}_n\left\{\varphi\left(Z;\hat{\psi},\hat{\mu}_t,\hat{\pi}_t\right)\right\} = 0$ gives:

$$\mathbb{P}_n\left\{\mathbf{1}(T=t)\frac{(Y-\hat{\mu}_t(X))}{\hat{\pi}_t(X)} + \hat{\mu}_t(X) - \hat{\psi}\right\} = 0 \Longleftrightarrow \hat{\psi} = \mathbb{P}_n\left\{\mathbf{1}(T=t)\frac{(Y-\hat{\mu}_t(X))}{\hat{\pi}_t(X)} + \hat{\mu}_t(X)\right\}$$

In this example, the estimating equation approach and the bias correction approach give the same estimator!

3.3 Targeted learning

Another alternative to one-step correction is to use targeted maximum likelihood estimation (TMLE) [13, 12]. TMLE actually uses a form of the plug-in estimator $\hat{\psi}_{pi}$ and does not correct bias on ψ the parameter scale by adding an estimate of bias to the plug-in estimator; instead, it aims to correct bias on the distributional scale from $\widehat{\mathbb{P}}$ to a fluctuated $\widehat{\mathbb{P}}^*$. More particularly, the TMLE estimator consists in most cases in estimating a nuisance parameter $\widehat{\eta}$ and then build a fluctuated nuisance estimate $\widehat{\eta}^*$ based on the efficient influence function φ so that we have:

$$\mathbb{P}_n\varphi\left(Z;\hat{\psi}_{pi},\hat{\eta}^*\right)=0 \tag{7}$$

Eventually, the targeted learning estimator applies the plug-in estimator to the fluctuated $\widehat{\eta}^*$ is:

$$\hat{\psi}^{tl}=\psi(\widehat{\mathbb{P}}^*)=\mathbb{P}_n\{\widehat{\eta}^*\}. \tag{8}$$

Exemple 9. *(Average treatment effect) Recall we have $\psi=\mathbb{E}\{\mu_t(X)\}$ and that:*

$$\varphi(Z;\mathbb{P})=\varphi\left(Z;\psi,\mu_t,\pi\right)=\mathbf{1}(T=t)\frac{(Y-\mu_t(X))}{\pi_t(X)}+\mu_t(X)-\psi$$

Our goal is to estimate $\hat{\mu}_t$ and $\hat{\pi}_t$ and then construct an optimized parameter $\hat{\mu}_t^*$ over a fluctuation so that (7) writes as:

$$\mathbb{P}_n\left\{\mathbf{1}(T=t)\frac{(Y-\hat{\mu}_t^*(X))}{\hat{\pi}_t(X)}+\hat{\mu}_t^*(X)-\hat{\psi}\right\}=0$$

For any nuisance parameters $\bar{\mu}_t$, $\bar{\pi}_t$, the $\bar{\pi}_t$-specific fluctuation model we consider for $\bar{\mu}_t$ is

$$\mathcal{M}(\bar{\mu}_t,\bar{\pi}_t)=\left\{(x,t)\mapsto\bar{\mu}_t^{\epsilon}(x)=\operatorname{expit}\left(\operatorname{logit}(\bar{\mu}_t(x))+\epsilon\frac{\mathbf{1}(T=t)}{\bar{\pi}_t(x)}\right):t\in\mathbb{R}\right\}\subset\mathcal{M} \tag{9}$$

The fluctuation $\mathcal{M}(\bar{\mu}_t,\bar{\pi}_t)$ is a one-dimensional parametric model indexed by the real-valued parameter ϵ that goes through $\bar{\mu}_t$ (at $\epsilon=0$). In targeted learning, the goal is to build an estimator of μ_t better than $\hat{\mu}_t$. We seek this estimator in the fluctuation $\mathcal{M}(\hat{\mu}_t,\hat{\pi}_t)$, where $\hat{\pi}_t$ is an estimator of π_t bounded away from 0 and 1. The estimator is written as $\hat{\mu}_t^{\epsilon_n}$ for some data-driven $\epsilon_n\in\mathbb{R}$. To evaluate the candidates, we use the empirical risk function $\epsilon\mapsto\mathbb{P}_n(\mathcal{L}(\hat{\mu}_t^{\epsilon}))$, where the loss function $\mathcal{L}$ is given by

$$-\mathcal{L}(\mu_t)=Y\log\mu_t(X)+(1-Y)\log(1-\mu_t(X))$$

The oracle risk $\mathbb{P}\mathcal{L}(\bar{\mu}_t)$ measures the discrepancy between $\bar{\mu}_t$ and the true μ_t. Importantly, note that the true function μ_t minimizes $\bar{\mu}_t\mapsto\mathbb{P}\mathcal{L}(\bar{\mu}_t)$ over all measurable functions, and that $\epsilon^*\in\mathbb{R}$ minimizing $\epsilon\mapsto\mathbb{P}\mathcal{L}(\hat{\mu}_t^{\epsilon})$ actually identifies the closest element in the fluctuation $\mathcal{M}(\hat{\mu}_t,\hat{\pi}_t)$ to μ_t. In practice, ϵ_n is chosen as the minimizer of the empirical risk function.

$$\epsilon_n=\underset{\epsilon\in\mathbb{R}}{\arg\min}\mathbb{P}_n\left(\mathcal{L}\left(\hat{\mu}_t^{\epsilon}\right)\right)$$

Define $\hat{\mu}_t^*\doteq\hat{\mu}_t^{\epsilon_n}$. We justify the impact of transitioning from $\hat{\mu}_t$ to $\hat{\mu}_t^*$ along the fluctuation $\mathcal{M}(\hat{\mu}_t,\hat{\pi}_t)$ in two steps. First, there is no need to iterate the updating procedure. Even if $\hat{\mu}_t^*$ were further fluctuated along $\mathcal{M}(\hat{\mu}_t^*,\hat{\pi}_t)=\{\hat{\mu}_t^{*,\epsilon'}:\epsilon'\in\mathbb{R}\}$, no change would occur. This is due to the one-to-one smooth correspondence between ϵ' and ϵ (namely, $\epsilon'=\epsilon+\epsilon_n$). Consequently, the derivative of $\epsilon\mapsto\mathbb{P}_n(\mathcal{L}(\hat{\mu}_t^{\epsilon}))$ at ϵ_n is 0, implying $\frac{d}{d\epsilon}\mathbb{P}_n(\mathcal{L}(\hat{\mu}_t^{*,\epsilon}))|_{\epsilon=0}=0$. Moreover, we show in Appendix 5 that:

$$\frac{d}{d\epsilon}\,\mathbb{P}_n\left(\mathcal{L}\left(\mu_t^{*,\epsilon}\right)\right)\big|_{\epsilon=0}=-\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{1}(T_i=t)}{\hat{\pi}_t(X_i)}\left(Y_i-\hat{\mu}_t^*(X_i)\right) \tag{10}$$

Therefore, this equivalently writes as:

$$\mathbb{P}_n \left\{ \mathbf{1}(T=t) \frac{(Y - \hat{\mu}_t^*(X))}{\hat{\pi}_t(X)} \right\} = 0 \tag{11}$$

Likewise, by definition of the targeted learning estimator in (8)*:*

$$\mathbb{P}_n \left\{ \hat{\mu}_t^*(X) - \hat{\psi} \right\} = 0 \tag{12}$$

We have therefore proven that the targeted learning estimator ψ^{tl} along with the fluctuated $\hat{\mu}_t^*$ solve the efficient influence curve equation in (7)*.*

Despite its asymptotic equivalence to the one-step estimator, an argument for using TMLE is that it could give better finite-sample properties, for example if $\psi(\mathbb{P})$ and $\psi(\widehat{\mathbb{P}}^*)$ are bounded, e.g., in $[-1, 1]$. A simple one-step estimator can potentially lie outside such bounds on the parameter space depending on the behavior of the correction term $\mathbb{P}_n\{\varphi(z; \widehat{\mathbb{P}})\}$ [13, 6].

4 Asymptotical analysis of the one-step estimator

The one-step estimator can be analyzed in some generality, as it is a simple average of an estimated function. By definition we have the important decomposition

$$\begin{aligned}
\widehat{\psi}^{os} - \psi &= \psi(\widehat{\mathbb{P}}) + \mathbb{P}_n\{\varphi(Z; \widehat{\mathbb{P}})\} - \psi(\mathbb{P}) \\
&= (\mathbb{P}_n - \mathbb{P}) \{\varphi(Z; \widehat{\mathbb{P}})\} + R_2(\widehat{\mathbb{P}}, \mathbb{P}) \\
&= (\mathbb{P}_n - \mathbb{P}) \{\varphi(Z; \mathbb{P})\} + (\mathbb{P}_n - \mathbb{P}) \{\varphi(Z; \widehat{\mathbb{P}}) - \varphi(Z; \mathbb{P})\} + R_2(\widehat{\mathbb{P}}, \mathbb{P})
\end{aligned}$$

To construct a $\sqrt{n}$ consistent estimator. We will assign specific names to each of these terms and examine them in detail, one at a time:

- Sample average: $S^* = (\mathbb{P}_n - \mathbb{P}) \{\varphi(Z; \mathbb{P})\}$
- Empirical Process: $(\mathbb{P}_n - \mathbb{P}) \left\{ \varphi(Z; \widehat{\mathbb{P}}) - \varphi\left(Z; \mathbb{P}\right) \right\}$
- Second-Order Remainder: $R_2\left(\widehat{\mathbb{P}}, \mathbb{P}\right) = \left[\psi(\widehat{\mathbb{P}}) - \psi\left(\mathbb{P}\right)\right] - \mathbb{P}\left\{\varphi\left(Z; \widehat{\mathbb{P}}\right)\right\}$

The first term

$$(\mathbb{P}_n - \mathbb{P}) \{\varphi(Z; \mathbb{P})\}$$

is a simple sample average of a fixed function, and so by the central limit theorem, for example, it behaves as a normally distributed random variable with variance $\text{var}(\varphi)/n$, up to error $o_{\mathbb{P}}(1/\sqrt{n})$.

Empirical Process The term $(\mathbb{P}_n - \mathbb{P}) \left\{ \varphi(Z; \widehat{\mathbb{P}}) - \varphi\left(Z; \mathbb{P}\right) \right\}$ is called empirical processes because it is a special kind of stochastic processes based on the empirical distribution $\widehat{\mathbb{P}}$. There are two ways we can make sure this term is $o_{\mathbb{P}}(1/\sqrt{n})$: assuming the estimators are not too complex (Donsker classes) or using cross-fitting. We'll discuss what these are shortly but first we need to define a consistency characteristic on the estimated influence function.

Definition 4.1. *Let φ be a influence function, we say that an estimator of the influence function $\hat{\varphi}$ is L_2-consistent if*

$$\|\hat{\varphi}(Z) - \varphi(Z)\|_2 \to_{\mathbb{P}} 0$$

where the norm is the L_2 norm $\|f\|^2 = \int f^2 dp$.

We now assume that our influence function is φ is L_2-consistent, which is necessary for both cross-fitting and Donsker classes.

Intuitively, when estimating $\widehat{\mathbb{P}}$ using the same sample on which $\mathbb{P}_n$ operates, the one-step estimator's bias correction involves "double-dipping" — using the same data for two distinct tasks. The first task is constructing the nuisance components in $\widehat{\mathbb{P}}$, and the second is estimating the bias term $\mathbb{P}_n\{\varphi(Z;\widehat{\mathbb{P}})\}$. This double use of data can lead to overfitting [3].

Donsker Conditions One natural way to avoid overfitting in general is to not fit overly complex models. Donsker classes encompass a variety of function spaces, including smooth parametric models, bounded monotone functions, smooth functions with bounded partial derivatives, Sobolev classes, and functions with bounded sectional variation, among others. In particular, Vaart [11] proves:

Proposition 10. *Let φ be a influence function and $\hat{\varphi}$ a Donsker and L_2-consistent estimated influence function, then*

$$(\mathbb{P}_n - \mathbb{P})\left\{\varphi(Z;\widehat{\mathbb{P}}) - \varphi\left(Z;\mathbb{P}\right)\right\} = o_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right)$$

Cross-fitting A straightforward way to eliminate this term is by using cross-fitting, also known as sample splitting. This involves estimating all components of $\mathbb{P}$ involved in the influence function with one sample and evaluating the estimator on a separate sample. Similar to cross-validation, this method can be extended to multiple K-folds. Here, $\hat{\varphi}^{(-k)}$ represents the influence function estimated excluding the kth fold, and I_k denotes the set of indices included in the kth fold. Kennedy [6], Fisher and Kennedy [4] provide the following result.

Proposition 11. *Let φ be a influence function and $\hat{\varphi}$ a L_2-consistent estimated influence function. We denote $\mathcal{I} = \{1, \cdots, n\}$, let $\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_K$ be a partition of $\mathcal{I}$. We define $\mathcal{I}_{-k} = \mathcal{I} \backslash \mathcal{I}_k$, then*

$$\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{I}_k}\left(\mathbb{P}_n^{(k)} - \mathbb{P}\right)\left(\varphi(Z,\widehat{\mathbb{P}}^{(-k)}) - \varphi\left(Z,\mathbb{P}\right)\right) = o_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right)$$

where $\mathbb{P}_n^{(k)}$ is the empirical measure and the fold $\mathcal{I}_k$ and where $\widehat{\mathbb{P}}_{-k}$ denote an estimator of $\mathbb{P}$ (or its relevant components appearing in the influence curve φ) that only uses observations $F_i \neq k$, i.e., excludes fold k.

The primary drawback of using cross-fitting to control the empirical process term is the potential for finite-sample bias due to effectively utilizing less data in each step. While this bias is negligible in large samples, it may be a concern in smaller studies.

Second-Order Remainder The term $R_2\left(\widehat{\mathbb{P}},\mathbb{P}\right) = \left[\psi(\widehat{\mathbb{P}}) - \psi\left(\mathbb{P}\right)\right] - \mathbb{P}\left\{\varphi\left(Z,\widehat{\mathbb{P}}\right)\right\}$ is known as the second-order remainder because it is involves second-order products of differences between $\mathbb{P}_n$ and $\mathbb{P}$. Moreover, like the empirical process term, it can usually be shown to disappear at a "second-order" rate like $o_{\mathbb{P}}(1/\sqrt{n})$, when each nuisance error is of the order $n^{-1/4}$, the product of will indeed be of the order $1/\sqrt{n}$. The difference is that these conditions are often specific to the exact parameter and model we're working with, unlike the empirical process term which we've seen can be bounded in a very general way.

Exemple 12. *(Average treatment effect) We have $\psi = \mathbb{E}\left\{\mu_t(X)\right\}$ and we have from the calculations above:*

$$\varphi(Z;\mathbb{P}) = \frac{\mathbf{1}(T=t)}{\mathbb{P}(T=t \mid X)}\left(Y - \mu_t(X)\right) + \mu_t(X) - \psi$$

Then, the remainder is given by

$$R_2(\bar{P},P) = \int\left\{\frac{1}{\bar{\pi}_t(x)} - \frac{1}{\pi_t(x)}\right\}\left\{\mu_t(x) - \bar{\mu}_t(x)\right\}\pi_t(x)dP(x)$$

where $\pi_t(x) = P(T = t \mid X = x)$ *and* $\bar{\pi}_t(x) = \bar{P}(T = t \mid X = x)$*, and similarly for* $\mu_t(x) = \mathbb{E}_P(Y \mid X = x, T = t)$*. Therefore if* $\widehat{\pi}_t(x) \geq \epsilon$ *with probability one we have*

$$|R_2(\mathbb{P}_n, \mathbb{P})| \leq \left(\frac{1}{\epsilon}\right) \int \left|\pi_t(x) - \widehat{\pi}_t(x)\right| \left|\mu_t(x) - \widehat{\mu}_t(x)\right| d\mathbb{P}(x) \leq \left(\frac{1}{\epsilon}\right) \left\|\widehat{\pi}_t - \pi_t\right\| \left\|\widehat{\mu}_t - \mu_t\right\|$$

by Cauchy-Schwarz. Thus if $\|\widehat{\pi}_t - \pi_t\| = o_{\mathbb{P}}\left(n^{-1/4}\right)$ *and* $\|\widehat{\mu}_t - \mu_t\| = o_{\mathbb{P}}\left(n^{-1/4}\right)$*, for example, then* $T_2 = o_{\mathbb{P}}(1/\sqrt{n})$*, as desired (though note that* π *and* μ *do not have to be estimated at the same rates for* $T_2 = o_{\mathbb{P}}(1/\sqrt{n})$ *to hold: any such combination whose product is* $o_{\mathbb{P}}(1/\sqrt{n})$ *would suffice). Conditions under which* $L_2(\mathbb{P})$ *errors satisfy, e.g.,* $\|\widehat{\pi}_t - \pi_t\| = o_{\mathbb{P}}\left(n^{-1/4}\right)$ *are available for many popular estimators.*

Summary In summary, when the empirical process and the remainder terms are of the order $o_{\mathbb{P}}(1/\sqrt{n})$, then the sample average term S^* dominates the decomposition, and so

$$\sqrt{n}(\widehat{\psi}^{os} - \psi) = \sqrt{n}S^* + o_{\mathbb{P}}(1) \rightsquigarrow N(0, \text{var}\{\varphi(Z; \mathbb{P})\})$$

by the central limit theorem and Slutsky's theorem. The conclusion provides key insights: $\widehat{\psi}^{os}$ is root-n consistent, and it is asymptotically normal. This implies that asymptotically valid 95% confidence intervals for ψ can be constructed using the expression $\widehat{\psi}^{os} \pm 1.96\sqrt{\widehat{\text{var}}\{\varphi(Z; \widehat{\mathbb{P}})\}/n}$.

5 Local Minimax Optimality in Nonparametric Efficiency Bounds

After having built an estimator for the target parameter ψ, a next line of business is to understand lower bounds or benchmarks for estimation error. In other words, how well can we possibly hope to estimate the parameter ψ over the model $\mathcal{P}$? This question tackles two problems: (i) defining a benchmark that no estimator can improve upon, and (ii) finding an estimator that in fact attains the benchmark.

A standard way to benchmark estimation error is through minimax lower bounds of the form

$$\inf_{\widehat{\psi}} \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[\{\widehat{\psi} - \psi(P)\}^2\right] \geq R_n$$

These kinds of lower bounds say that the risk for estimating ψ (in this case, in terms of worstcase mean squared error), over the model $\mathcal{P}$, cannot be smaller than R_n. In the litterature, a classic benchmarking or lower bound result for smooth parametric models uses the Cramer-Rao bound [2, 14]. In a more subtle form, for smooth parametric models $\mathcal{P} = \{P_\theta : \theta \in \mathbb{R}\}$ and smooth functionals (i.e., with P_θ and $\psi(\theta)$ differentiable in θ), Van der Vaart [15] states the following.

Proposition 13. *Assume* P_θ *is differentiable in quadratic mean at* θ *with nonsingular Fisher information* $I_\theta = \text{var}_\theta\{s_\theta(Z)\}$*. If* $\psi(\theta)$ *is differentiable at* θ*, with* $\psi'(\theta) = \frac{\partial}{\partial\theta}\psi(\theta)$*, then for any estimator* $\widehat{\psi}$ *it follows that*

$$\inf_{\delta>0} \liminf_{n\to\infty} \sup_{\|\theta'-\theta\|<\delta} n\mathbb{E}_{\theta'} \left[\left\{\widehat{\psi} - \psi(\theta')\right\}^2\right] \geq \psi'(\theta) \,\text{var}_\theta\{s_\theta(Z)\}^{-1} \psi'(\theta)^{\mathrm{T}}$$

where $s_\theta(z) = \frac{\partial}{\partial\theta} \log p_\theta(z)$ is the score function, i.e. the (asymptotic, worst-case) mean squared error cannot be smaller than $\psi'(\theta)^2/n \,\text{var}_\theta\{s_\theta(Z)\}$, for any estimator $\widehat{\psi}$ in a smooth parametric model.

However, what if anything does this say about larger semi- or nonparametric models? Can the above Cramer-Rao bounds be exploited to construct lower bound benchmarks there as well? These questions will be answered in the following .

To answer the previous question, take the parametric submodel (3), and first note that for this submodel the score function is

$$\frac{\partial}{\partial\epsilon} \log p_\epsilon(z)\Big|_{\epsilon=0} = \frac{\partial}{\partial\epsilon} \log\{1 + \epsilon h(z)\}\Big|_{\epsilon=0} = h(z).$$

In that case, the Cramer-Rao lower bound for the associated P_ϵ is given by

$$\frac{\psi'(P_\epsilon)^2}{\mathrm{var}_{P_\epsilon}\{s_\epsilon(Z)\}} = \frac{\left(\frac{\partial}{\partial\epsilon}\psi(P_\epsilon)\big|_{\epsilon=0}\right)^2}{\mathbb{E}_{P_\epsilon}\{h(Z)^2\}}$$

Since for this submodel, the score is $s_\epsilon(z) = h(z)$. Using pathwise differentiability (5) we have

$$\frac{\partial}{\partial\epsilon}\psi(P_\epsilon)\bigg|_{\epsilon=0} = \int \varphi(z;\mathbb{P})h(z)d\mathbb{P}(z).$$

Therefore, over all Cramer-Rao bounds at $\epsilon = 0$ we have

$$\sup_{P_\epsilon}\frac{\psi'(P_\epsilon)^2}{\mathrm{var}_{P_\epsilon}\{s_\epsilon(Z)\}} = \sup_h \frac{\mathbb{E}\{\varphi(Z;\mathbb{P})h(Z)\}^2}{\mathbb{E}\{h(Z)^2\}} \leq \mathbb{E}\left\{\varphi(Z;\mathbb{P})^2\right\} = \mathrm{var}\{\varphi(Z)\}$$

where the first equality follows by pathwise differentiability and the form of the submodel, and the inequality by Cauchy-Schwarz. The fact that the greatest lower bound is not just bounded above by $\mathbb{E}\left(\varphi^2\right) = \mathrm{var}(\varphi)$, but that this upper bound is actually attained follows from the fact that, for one of the submodels we can take $h(z) = \varphi(z;\mathbb{P})$, as long as φ is in the tangent space (i.e., closure of submodel score space)

Remark 14. *In proper nonparametric models, the tangent space encompasses the entire Hilbert space of mean-zero, finite-variance functions, allowing only one influence curve φ, which also serves as the efficient influence function [6]. In semiparametric models with restricted tangent spaces, multiple influence curves may satisfy the expansion, but only the one that is also a valid score is the efficient influence function.*

Therefore the variance of the efficient influence function

$$\mathrm{var}\{\varphi(Z;\mathbb{P})\} \tag{13}$$

acts as a nonparametric analog of the Cramer-Rao bound. This is critically important, as it implies no estimator can have smaller mean squared error than (13), in a local asymptotic minimax sense. In particular, since we have shown for the one-step estimator $\widehat{\psi}^{os}$ that

$$\sqrt{n}(\widehat{\psi}^{os} - \psi) \rightsquigarrow N(0, \mathrm{var}\{\varphi(Z)\})$$

then we can say the one-step estimator attains the nonparametric efficiency bound.

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Appendix

This contains additional details for the explanations in the main text.

We prove the following result:

$$\frac{d}{d\epsilon}\,\mathbb{P}_n\left(\mathcal{L}\left(\mu_t^{*,\epsilon}\right)\right)\big|_{\epsilon=0} = -\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{1}(T_i=t)}{\hat{\pi}_t(X_i)}\left(Y_i-\hat{\mu}_t^*(X_i)\right) \tag{14}$$

Proof.

$$\begin{aligned}
\frac{d}{d\epsilon}\,\mathbb{P}_n\left(\mathcal{L}\left(\mu_t^{*,\epsilon}\right)\right)\big|_{\epsilon=0} &= -\frac{1}{n}\frac{d}{d\epsilon}\sum_{i=1}^{n}\left(Y_i\log\mu_t^{*,\epsilon}(X_i)+(1-Y_i)\log\left(1-\mu_t^{*,\epsilon}(X_i)\right)\right)\Big|_{\epsilon=0}\\
&= -\frac{1}{n}\sum_{i=1}^{n}\left(\frac{Y_i}{\mu_t^{*,\epsilon}(X_i)}-\frac{1-Y_i}{1-\mu_t^{*,\epsilon}(X_i)}\right)\times\frac{d}{d\epsilon}\mu_t^{*,\epsilon}(X_i)\Big|_{\epsilon=0}\\
&= -\frac{1}{n}\sum_{i=1}^{n}\frac{Y_i-\mu_t^{*,\epsilon}(X_i)}{\mu_t^{*,\epsilon}(X_i)\times\left(1-\mu_t^{*,\epsilon}(X_i)\right)}\frac{d}{d\epsilon}\mu_t^{*,\epsilon}(X_i)\Big|_{\epsilon=0}\\
&= -\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{1}(T_i=t)}{\pi_t(X_i)}\left(Y_i-\mu_t^{*,\epsilon}(X_i)\right)
\end{aligned}$$

where the last equality is derived from the following. Set $\alpha,\beta\in\mathbb{R}$. The derivative of $\epsilon\mapsto\operatorname{expit}(\alpha+\beta\epsilon)$ evaluated at $\epsilon=0$ satisfies

$$\frac{d}{d\epsilon}\operatorname{expit}(\alpha+\beta\epsilon)\Big|_{\epsilon=0} = \beta\operatorname{expit}(\alpha)(1-\operatorname{expit}(\alpha)).$$

Therefore, for any $(x,t)\in[0,1]\times\{0,1\}$,

$$\begin{aligned}
\frac{d}{d\epsilon}\mu_t^{*,\epsilon}\Big|_{\epsilon=0} &= \frac{\mathbf{1}(T=t)}{\pi_t(x)}\operatorname{expit}(\operatorname{logit}(\mu_t^{*,\epsilon}(x))[1-\operatorname{expit}(\operatorname{logit}(\mu_t^{*,\epsilon}(x))]\\
&= \frac{\mathbf{1}(T=t)}{\pi_t(x)}\mu_t^{*,\epsilon}(x)[1-\mu_t^{*,\epsilon}(x)].
\end{aligned}$$

Furthermore the same derivations that led to this also imply, that

$$\frac{d}{d\epsilon}\mathcal{L}\left(\bar{\mu}_t^{\epsilon}\right)(O)\Big|_{\epsilon=0} = \frac{\mathbf{1}(T=t)}{\bar{\pi}_t(X)}(Y-\bar{\mu}_t^{\epsilon}(X))$$

In this light, we can think of $\mathcal{M}(\bar{\mu}_t,\bar{\pi}_t)$ as a fluctuation of $\mathcal{M}$ in the direction of

$$(x,t,y)\mapsto\frac{\mathbf{1}(T=t)}{\bar{\pi}_t(x)}(y-\bar{\mu}_t(x))$$

which is a fluctuation of $\bar{M}$ in the direction of the first component of the efficient influence curve defined in (7). □